SMART LIBRARY REQUEST WORKFLOW IN SERVICENOW

🔷 Introduction

In many educational institutions, library activities are still handled manually, which creates several challenges. Students often find it difficult to know whether a required book is available or already issued. At the same time, librarians need to maintain records and process requests manually, which increases workload and chances of mistakes.

To overcome these issues, a digital solution is required that can simplify and automate the entire process. The Smart Library Request Workflow is designed using ServiceNow to provide an efficient and structured system for managing book requests.

This system allows students to submit requests online and enables librarians to review and process them through an automated workflow. It improves transparency, reduces manual effort, and ensures accurate record management. By implementing this system, library operations become faster, more organized, and user-friendly.

🔷 1) Identification Phase

Introduction

The identification phase is the starting point of the project, where the main problem is clearly defined. It helps in analysing the existing system and identifying its limitations. In traditional library systems, most tasks such as book issuing, request handling, and record maintenance are done manually.

Due to this, students face difficulty in checking book availability, and librarians spend more time managing records. There is also a higher risk of data inconsistency and delays in processing requests.

To address these challenges, the Smart Library Request Workflow is proposed. This system uses automation to handle book requests, approvals, and tracking. It provides a structured approach where students and librarians can interact through a centralized platform. The solution improves efficiency and reduces errors in the system.

🔷 2) Requirement Analysis

Introduction

Requirement analysis is the phase where system needs are collected and analyzed. It defines what the system should do and how it should perform. This step ensures that the final system meets user expectations and functions effectively.

In this project, the focus is on automating book request handling, maintaining book records, and enabling smooth communication between students and librarians.

Functional Requirements

- Students must be able to submit book requests
- Librarians should review and take action on requests
- The system must update book status automatically
- Users should be able to check request status
- Notifications should be sent for approvals or rejections
- Reports such as frequently borrowed books should be generated

Non-Functional Requirements

- The system must be secure and reliable
- It should support role-based access control
- The interface should be simple and easy to use
- The system should provide fast response time
- Data should be accurate and updated in real-time

Hardware Requirements

- Computer or Laptop
- Minimum 4GB RAM
- Stable Internet Connection

Software Requirements

- Platform: ServiceNow
- Operating System: Windows / Linux
- Browser: Google Chrome or Microsoft Edge

🔷 3) Project Planning Phase

Introduction

Project planning involves organizing tasks and defining the steps required to complete the project. It helps in proper execution and ensures that the project is completed within the given time frame.

In this phase, activities such as defining roles, designing workflows, and scheduling development tasks are planned.

Project Goals

- To automate the library book request system
- To reduce manual work and human errors
- To improve tracking and record management
- To enhance user experience for students and librarians

Project Milestones

- Role creation (Student, Librarian)
- Table creation (Book, Borrow Request)
- Field configuration
- Workflow automation setup
- Access control implementation
- Testing and reporting

Project Timeline

- Week 1: Requirement analysis
- Week 2: Planning and design
- Week 3: Table creation and configuration
- Week 4: Workflow automation
- Week 5: Security and reports
- Week 6: Testing and final demonstration

Risk Management

- Configuration mistakes → solved through testing
- Data errors → validation and verification
- Workflow failure → debugging and monitoring
- System downtime → proper maintenance

🔷 4) Development Phase

Introduction

The development phase focuses on building the system using the tools available in ServiceNow. It includes creating database tables, configuring fields, and implementing workflows.

Table Creation

Two main tables are developed to store system data.

Book Table

This table contains details of books such as:

- Book Title
- Author Name
- ISBN Number
- Availability Status

Borrow Request Table

This table maintains request-related details such as:

- Student Name
- Requested Book
- Request Date
- Approval Status

Field Configuration

Fields are created with appropriate data types like text, date, and reference. Status values include Available, Requested, Approved, Issued, Rejected, and Returned.

Workflow Automation

Using Flow Designer:

- Student submits request
- Request is forwarded to librarian
- Librarian approves/rejects
- System updates status automatically
- Notification is sent to the user

🔷 5) Project Design Phase

Introduction

The design phase defines the structure and layout of the system. It ensures that the system is well-organized and easy to use.

System Design

The system is based on two main roles:

- Student: submits requests
- Librarian: manages approvals and records

Database Design

The database includes:

- Book table
- Borrow request table

Both tables are linked using reference fields to maintain relationships.

User Interface Design

The interface is designed for simplicity:

- Easy-to-use request forms for students
- Dashboard view for librarians
- Clear status tracking system

🔷 6) Project Documentation

Introduction

Documentation explains how the project is developed and how it works. It is useful for understanding and maintaining the system in the future.

Purpose of Documentation

- To describe system functionality
- To guide users in operating the system
- To support future improvements

Technical Documentation

Includes details about:

- Table structure
- Workflow configuration
- Access control rules
- Platform features of ServiceNow

User Documentation

Provides instructions such as:

- How to login
- How to request books
- How to approve requests
- How to check status

🔷 7) Project Demonstration

Introduction

The demonstration phase shows how the system works in real-time and verifies that all features are functioning correctly.

Demonstration Process

1. Student logs into the system
2. Submits a request for a book
3. Request is sent to librarian
4. Librarian reviews and approves/rejects
5. System updates book and request status
6. Student receives notification

Outcome

The system successfully automates the library management process. It reduces manual effort, improves accuracy, and provides a faster way to handle book requests. The project demonstrates how workflow automation can improve traditional systems.

🔷 Conclusion

The Smart Library Request Workflow developed using ServiceNow provides an effective solution to replace manual library processes with an automated system. It allows students to request books easily and enables librarians to manage approvals efficiently through workflows. This reduces time consumption, minimizes errors, and improves the overall management of library operations.

The project enhances transparency and user experience by offering features like real-time status tracking, notifications, and role-based access control. It ensures accurate record maintenance and smooth communication between users. Overall, the system demonstrates how automation can improve efficiency and can be further expanded with advanced features in the future.

🔷 STEPS FOR IMPLEMENTATION

1. Create Roles

Begin by defining user roles in the system to manage access control.
Create two roles:

- **Student** – can request books and view status
- **Librarian** – can manage books and approve requests

These roles ensure that each user has specific permissions based on their responsibilities.

2. Create Tables

Next, create custom tables to store system data:

- **Book Table (u_ book)** – maintains all book-related details
- **Borrow Request Table (u_ borrow_ request)** – stores information about book requests made by students

These tables act as the foundation of the application.

3. Add Fields to Book Table

Define necessary fields in the Book table to capture book information:

- **Title** – String (book name)
- **Author** – String (author name)
- **ISBN** – String (unique identifier)
- **Status** – Choice

Status options include:

- Available (default)
- Issued
- Lost

4. Add Fields to Borrow Request Table

Add required fields to manage request details:

- **Requested By** – Reference (User table)
- **Book** – Reference (Book table)
- **Request Date** – Date/Time
- **Status** – Choice

Options include:

- Requested (default)
- Approved
- Rejected
- Returned

These fields help track each request clearly.

5. Configure Reference Qualifier

Apply a reference condition on the **Book field** in the Borrow Request table.

- Condition: **Status = Available**

This ensures that users can only select books that are currently available, avoiding invalid requests.

6. Create Flow in Flow Designer

Develop an automated workflow using Flow Designer.

Flow Name: Borrow Request Approval Flow

Steps involved:

- Trigger flow when a request is created or modified
- Send approval request to the librarian

If approved:

- Update Book Status → Issued
- Update Request Status → Approved
- Notify the student via email

If rejected:

- Update Request Status → Rejected
- Send notification to the student

This automation reduces manual work and speeds up the process.

7. Create UI Policy

Create a UI Policy for the Borrow Request table to control form behavior.

- **Condition:** When Status = Issued
- **Action:** Make "Return Date" field mandatory

This ensures that important information is not missed during book return.

8. Configure Access Control (ACL)

Set permissions to secure the system based on roles.

Book Table

- Student → Read access
- Librarian → Full access (Read, Write, Delete)

Borrow Request Table

- Student → Create and Read
- Librarian → Read, Write, Delete

This ensures proper data security and controlled access.

9. Create Report

Generate a report to analyse system usage.

- **Report Name:** Most Borrowed Books
- **Table:** Borrow Request
- **Group By:** Book
- **Aggregation:** Count

This report helps identify popular books in the library.

10. Testing

Finally, test the system using sample data to ensure proper functionality:

1. Student submits a book request
2. Request is sent to the librarian
3. Librarian reviews and approves the request
4. Book status changes automatically to “Issued”
5. Student receives notification

Testing confirms that the workflow operates correctly and meets requirements.